

EXPERIMENT 8

Aim

To design and train a Convolutional Neural Network (CNN) using TensorFlow/Keras to classify images from the Fashion-MNIST dataset.

Problem Statement

Image classification is a key problem in computer vision.

The task is to correctly classify grayscale images of clothing items into their respective categories such as shirts, shoes, and bags using a CNN model.

Dataset Link

<https://www.kaggle.com/datasets/zalando-research/fashionmnist>

Methodology

Data Collection

The Fashion-MNIST dataset was loaded using Keras, containing grayscale images of clothing items such as shirts, shoes, and bags.

Data Preprocessing

Pixel values were normalized from the range **0–255 to 0–1** to improve model performance and training stability.

Data Reshaping

The dataset was reshaped into **28×28×1** format to match the input requirements of the CNN, where 1 represents the grayscale channel.

CNN Model Architecture

A Convolutional Neural Network was built consisting of:

- **Convolution Layers (ReLU)** → Extract important features like edges and textures
- **MaxPooling Layers** → Reduce spatial dimensions and computation
- **Flatten Layer** → Convert feature maps into a 1D vector
- **Fully Connected Dense Layer** → Learn complex patterns
- **Dropout Layer** → Prevent overfitting by randomly disabling neurons
- **Output Layer (Softmax)** → Classify images into multiple categories

Model Compilation

The model was compiled using the **Adam optimizer** for efficient learning and a **categorical loss function** suitable for multi-class classification.

Model Training

The model was trained for **10 epochs**, allowing it to learn patterns and improve classification accuracy.

Model Evaluation

The trained model was evaluated on the test dataset to measure its performance and generalization ability.

Result

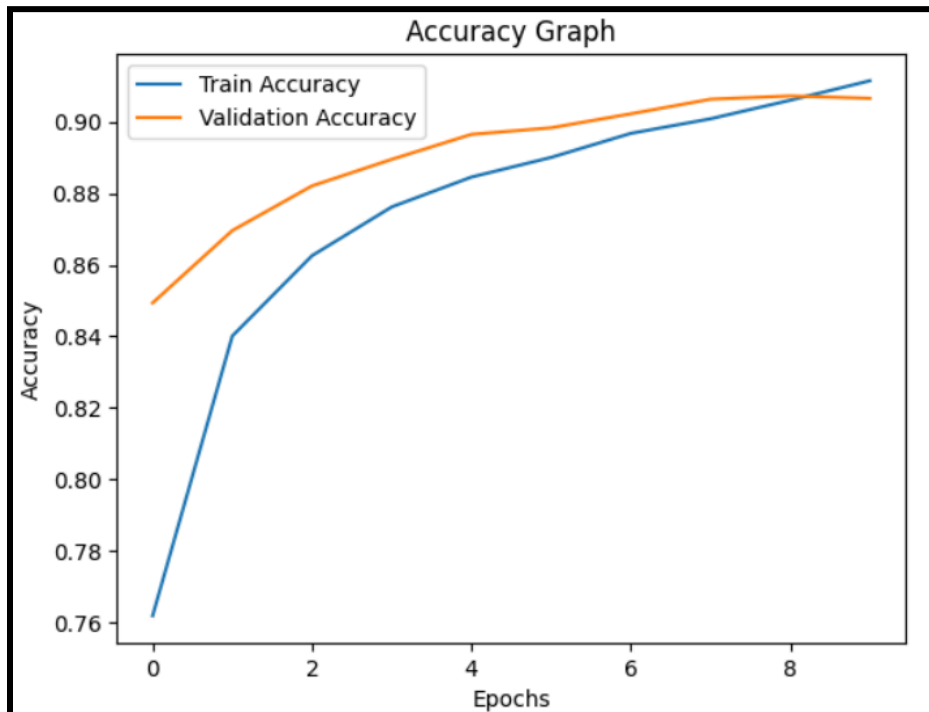
- **Test Accuracy:** ~90%
- **Loss:** Reduced over epochs

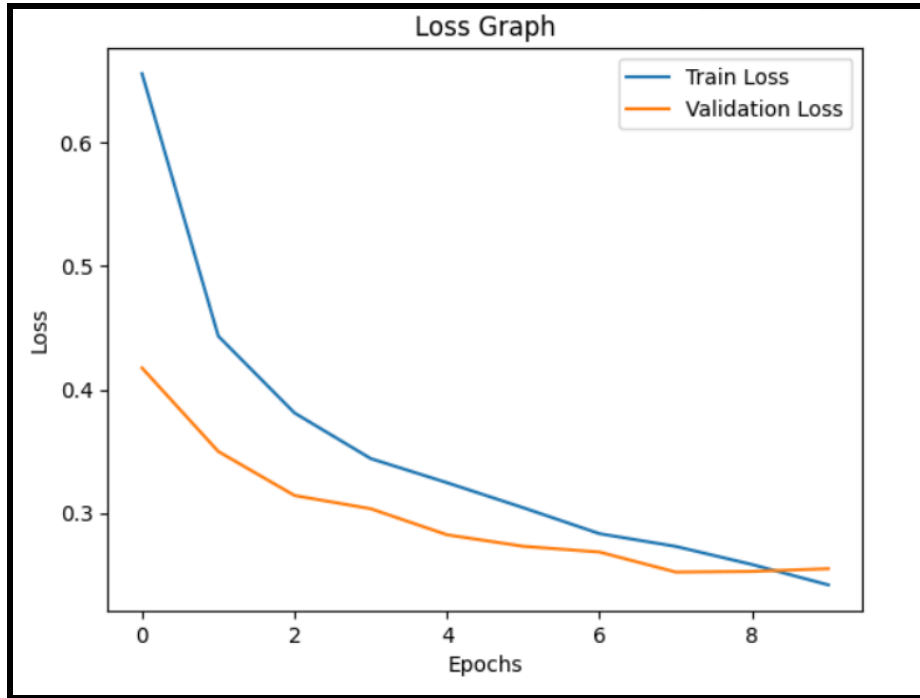
Performance Metrics:

- Accuracy
- Loss

Model Graphs:

- Accuracy vs Epochs
- Loss vs Epochs





Conclusion

The CNN model successfully classified images with high accuracy by learning important visual patterns directly from the input data. Through convolution operations, the model was able to automatically extract meaningful features such as edges, textures, and shapes, which significantly improved overall performance.

Additionally, the hierarchical feature extraction process allowed the model to capture both low-level and high-level patterns, making it highly effective for complex image recognition tasks. Due to its ability to learn spatial features efficiently, CNN is widely used and well-suited for various image classification applications.